

Psychotropic Drugs

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Psychotropic Drugs

- Locus of all mental activity is the brain
- Origin of psychiatric illness caused by many factors:
 - Genetics
 - Neurodevelopment factors
 - Drugs
 - Infections
 - Psychosocial experiences, etc.



- Theories behind use of psychotropic drugs focuses on neurotransmitters and their receptors
- Psychotropic drugs act by modulating neurotransmitters



Review: Cellular composition of brain

- **Neurons**-nerve cells that conduct electrical impulses
- **Neurotransmitter**-chemical that is released in response to an electrical impulse (**neuromessenger**).
 - Attaches to a receptors on cell surface and either inhibits or excites
 - Major target of psychotropic drugs

Therapeutic Effects of Psychotropic Meds

- Do not “cure” in some Dis
- Relieve or decrease symptoms
- Prevent or delay return of S/S
- Cannot be used as the sole tx for disorders
- Need informed consent before starting
- Are broad spectrum and have effects on a large number of S/S.
- Initial effects are sedative in nature
- May take weeks for effects to be seen



Reasons for Nonadherence:

- Meds are expensive
- Unpleasant side effects
- Feel better and decide no longer need
- Stigma associated with having a mental illness and taking meds
- Paranoia or fears about med usage



Efficacy of Psychotropics with Children & Elderly

- Use with great caution
- Start low and go slow for both elders and children!!
- Elders have decrease liver & renal function
- Risk of injuries and falls with elderly

Client & Family Teaching



Psychotropic Meds Classifications:

- Antipsychotics (neuroleptics)
- Mood Stabilizers
- Antidepressants
- Anxiolytics (antianxiety)
- Sedatives
- Hypnotics
- Psychostimulants
- Antihistamines, antimuscarinics, dopamine agonists

Indications for Antipsychotics

- The primary indication for this group of drugs is the presence of psychosis in such disparate disorders as:
 - 1-Schizophrenia
 - 2-Schizophreniform disorder
 - 3-Schizoaffective disorder
 - 4-Delusional disorder
 - 5-Brief psychotic disorder
 - 6-Psychoses secondary to a nonpsychiatric medical condition
 - 7-Mood disorders with mood-congruent or incongruent psychotic symptoms
 - 8-Mania
 - 9-Borderline personality disorder
 - 10-Substance-induced psychotic disorder
 - 11-Delirium and dementia
 - 12- Pervasive developmental disorder
 - 13-Tourette's syndrome
 - 14-Huntington's disease

Classical

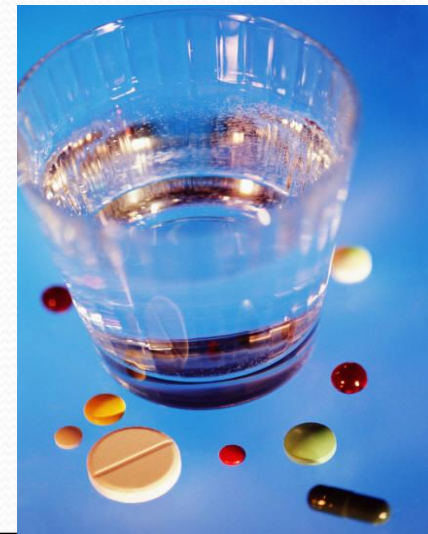
- Chlorpromazine
- Promazine
- Triflupromazine
- Thioridazine
- Stelazine Trifluoperazine
- Fluphenazine
- Prochlorperazine
- Loxapine
- Haloperidol
- Droperidol

atypicals

- Clozapine
- Risperidone
- Olanzapine
- Quetiapine
- Ziprasidone
- Aripiprazole
- Pimozide
- Paliperidone
- Sertindole
- Zotepine

First Gen Antipsychotic Meds

- Block predominantly dopamine activity
 - little effect on serotonin
- High incidence of abnormal movements
(Also blocks acetylcholine, norepinephrine to some degree)
- Blocks the H receptor for histamine
 - results in sedation and weight gain



Atypical Antipsychotics

- Action:
 - Blocks serotonin and to a lesser degree, dopamine receptors
 - Also block receptors for norepinephrine , histamine, acetylcholine



Atypical Antipsychotics- 2nd and 3rd generation drugs

- Nicer drugs and are used more!!
- Decrease positive and negative S/S of Schizophrenia
- These drugs block serotonin as well as dopamine
- Incidence of abnormal movements is lower!
- Biggest SE is wt. gain



Side effects

- **A-dopamine blockage effects:**

1-extrapyramidal side effects:

i-acathisia

ii-acute dystonia

iii-parkinsonian features

iiii-tardive dyskinesia

2-hyperprolactinemia

3-neuroleptic malignant syndrome

Neuroleptic Malignant Syndrome

- Rare, but fatal complication from all antipsychotic drugs
- See more with 1st gen drugs
- Severe muscle rigidity
- High temp up to 107
- Tachycardia
- Tachypnea
- Stupor
- Coma



● **B- anticholinergic side effects:**

- 1-blurred vision
- 2-urine retention
- 3-dry mouth
- 4-constipation
- 5-Some persons also have nausea and vomiting.
- 6-agitation©
- 7-halucination©

C- anti adrenergic side effects:

- 1-reflex tachycardia
- 2-postural hypotension
- 3-ECG change
- 4-sexual dysfunction
- 5-arythmia (VT) and sudden death

D- anti histaminic and 5H₁ blockade:

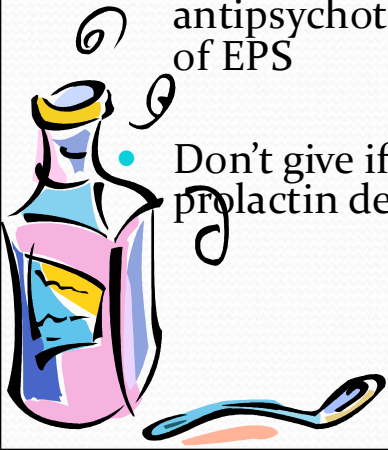
- 1-sedation
- 2-weight gain

● **E-Combined effects:**

- Seizure Threshold
- agitation; disorientation to time, person, and place; hallucinations
- A temporary leukopenia with a WBC count of about 3,500 is a common, but not serious problem
- Agranulocytosis
- Thrombocytopenic or nonthrombocytopenic purpura, hemolytic anemias, and pancytopenia may occur rarely
- Allergic dermatitis and photosensitivity can occur
- Urticarial, maculopapular, petechial, and edematous eruptions can occur early in treatment, generally in the first few weeks, and remit spontaneously (Persons should be warned of this adverse effect, spend no more than 30 to 60 minutes in the sun, and use sunscreens)
- retinal pigmentation
- Jaundice
- Elevations of liver enzymes

Contraindications of Traditional Antipsychotics (1st Gen):

- Blood dyscrasias
- Liver, renal, or cardiac insufficiency
- CNS depressants, including ETOH
- Tegretol in conjunction with antipsychotics causes up to 50% reduction in antipsychotic concentrations
- SSRI's in conjunction with antipsychotics may cause sudden onset of EPS
- Don't give if have: Parkinson's disease, prolactin dependent cancer of the breast
- Cigarette smoking causes reduced plasma concentrations of antipsychotics
- Luvox in conjunction with antipsychotics causes increased concentrations of Haldol and Clozaril
- Beta Blockers in conjunction with antipsychotics cause severe hypotension
- Antidepressants in conjunction with antipsychotics may cause increased antidepressant concentrations



Contraindications for Atypical Antipsychotics:

- Known hypersensitivity
- CNS depression, including ETOH
- Blood dyscrasias in clients with Parkinson's disease
- Liver, renal, or cardiac insufficiency
- Use with caution in diabetics, elderly, or debilitated
- SSRIs in conjunction with antipsychotics may cause sudden onset of EPS
- Cigarette smoking causes reduced plasma concentrations
- Tegretol (carbamazepine) in conjunction with antipsychotics causes up to 50% reduction in antipsychotic levels
- Luvox (fluvoxamine) in conjunction with antipsychotics causes increased concentrations of Haldol & Clozaril
- Beta Blockers in conjunction with antipsychotics cause severe hypotension
- Antidepressants in conjunction with antipsychotics may cause increased antidepressant concentrations



Antipsychotics

- Can be given be given as an IM injection (depot preparations) if have difficulty taking oral meds.
- Can use lower doses when given IM, so less risk of tardive dyskinesia

Mood Stabilizers

- Used in the treatment of Manic (Bipolar) disorder, and in some forms of depression
- Drugs used: **Lithium** and **Antiepileptic Drugs**



Lithium



- Mechanism of action unknown
- Interacts with sodium and K^+
- Alters electrical conductivity
 - potential threat to all body functions that are regulated by electrical currents
- Can cause polyuria and polydipsia due to Na and K alterations
- Has the lowest therapeutic index of all psych drugs
- Have to monitor blood levels of this drug

Lithium



- Maintenance blood levels of lithium are usually 0.4-1.3 mEq (toxicity occurs with levels > 1.5 mEq/L)
- Sign of toxicity is a fine intention tremor that becomes more pronounced and coarse.
- Risk of thyroid & kidney disease
- If toxic s/s occur discontinue the drug and notify health care provider
- Lithium should be taken with food
- Client must eat a balanced diet with normal sodium intake and take in adequate fluid (about 2-3 liters/day).
- Excretion is dependent on this.
- Dehydration and salt restriction can increase lithium levels & cause toxicity.
- Takes 2-3 weeks for lithium to become effective (may use antipsychotic until therapeutic levels are reached)

Signs & symptoms of lithium toxicity:

- Fine hand tremors that progress of coarse tremors
- Mild GI upset progressing to persistent upset
- Slurred speech and muscle weakness progressing to mental confusion
- **Severe Toxicity:**
 - decrease level of consciousness to stupor and finally coma
 - Seizures, severe hypotension, severe polyuria with dilute urine

Lithium:



- Lithium serum concentrations are increased by fluoxetine (Prozac), ACE inhibitors, diuretics, and NSAIDs
- Lithium serum concentrations are decreased by theophylline, osmotic diuretics, and urine alkalinizers

Contraindications for Lithium:

- Renal disease
- Cardiac disease
- Severe dehydration
- Sodium depletion
- Brain damage
- Pregnancy or lactation
- Use with caution in the elderly or clients with diabetics, thyroid disorders, urinary retention, and seizures



Anticonvulsants/Antiepileptic Drugs

- Causes an increase in GABA in the CNS-which causes a decrease in anxiety.
- Reduce the mood swings with bipolar



Anticonvulsants/Antiepileptic Drugs

- Tegretol (carbamazepine)-also used to treat severe pain (i.e. trigeminal neuralgia)
- Depakote (valproic acid)-can cause hepatic failure, pancreatitis, & thrombocytopenia.
Watch for liver failure
- Klonopin (clonazepam)
- Lamictal (Lamotrigine)-can have a rare but fatal dermatological condition

Toxic Effects of Anticonvulsants:

- Tegretol can cause agranulocytosis and aplastic anemia
- Depakote can cause liver dysfunction, hepatic failure, and blood dyscrasias including thrombocytopenia
- Depakote interacts with drugs that are hepatically metabolized

Contraindications for Anticonvulsants :

- Hepatic or renal disease
- Pregnancy
- Lactation
- Presence of blood dyscrasias



Unique teaching needs with anticonvulsants:

- Monitor blood levels of mood stabilizers to prevent toxicity
- Monitor liver, renal function tests and CBCs
- Depakote must be swallowed whole, not cut, chewed, or crushed to prevent irritation

Antidepressants

- Tx of depressive moods, including bipolar disease
- 4 categories:
 - Tricyclics
 - MAOI's
 - SSRI'S
 - Atypical Antidepressants



Antidepressant Drugs

- Tricyclics-

- Amitryptiline
- Clomipramine
- Dothiepin
- Imipramine
- Lofepramine
- Trimipramine
- Nortriptyline

- MAOI's-

Phenelzine
Isocarboxazid
tranylcypromine

- SSRI's-

- Citalopram and Escitalopram
- Flouxetine
- Fluvoxamine
- Paroxetine
- Sertraline
- duloxetine

Atypical Antidepressants

- Inhibits selective reuptake of serotonin:
Trazodone (desyrel)
- Norepinephrine Dopamine Reuptake Inhibitor (NDRI):
Wellbutrin (Bupropion)
- Serotonin Norepinephrine Reuptake Inhibitor-(SNRI):
venlafaxine, duloxetine
- Increases release of serotonin & norepinephrine : Remeron (mirtazapine)

Atypical Antidepressants

- **Trazodone**=
alternative to TCA's
 - Can cause orthostatic hypotension, sedation, & priapism in males
- **Remeron**= causes sedation, weight gain, dry mouth, constipation
- **Wellbutrin** (zyban)= rarely causes sedation, wt. Gain, or sexual dysfunction.
 - Used for smoking cessation. Most common S/E are headaches, insomnia & nausea
 - Can lower seizure threshold – causes seizures



Major Indications for Antidepressants

- Major Depressive disorder
- Bipolar depression
- Obsessive-Compulsive
- Anxiety
- Panic disorder
- PTSD
- Substance Abuse
- Chronic Pain
- Tourette's Disorder
- ADHD
- Eating disorders
- Sleep disorders
- Migraines
- Enuresis
- PMTS

Tricyclics:

- Blocks the reuptake of norepinephrine and serotonin
- Tricyclic drugs block the muscarine receptors (so anticholinergic effects)
- **Other side effects:**
 - orthostatic hypotension
 - sedation
 - wt. gain
 - confusion-esp. elderly
 - arrhythmias



Tricyclics Contraindications

- Dementia
- Suicidal clients
- Cardiac disease
- Pregnancy
- Seizure disorders
- Urinary retention
- Dose for elderly should be $\frac{1}{2}$ of adult dose
- TCA's and MAOIs are effective in tx. depression
 - are not as safe or as well tolerated as the newer antidepressants
- **Toxic Effects:**
possibility of cardiac toxicity and are toxic in overdose

SSRI's

- Are very safe and are not lethal in overdose
- Good choice with the elderly-very few side effects
- If used with MAOI's may cause Serotonin Syndrome=seizure, death
- If used with TCA's may cause TCA toxicity
- Takes 2 weeks to feel effects

MAOI's

- **Nardil, Parnate, Marplan**
- Inhibits MAO, thus interfering with breakdown of norepinephrine, dopamine, and serotonin
- Toxic effects=
 - hypertensive crises
- Avoid foods with tyramine (aged cheese, red wine, beer, chocolate, etc.)
- MAOI's don't play well with other drugs!!



Antianxiety/Anxiolytic Drugs

- GABA exerts an inhibitory effect on neurons
- These drugs enhance this effect and produce a sedative effect
- Therefore reduce anxiety
- The most common used drugs here are the **Benzodiazepines**



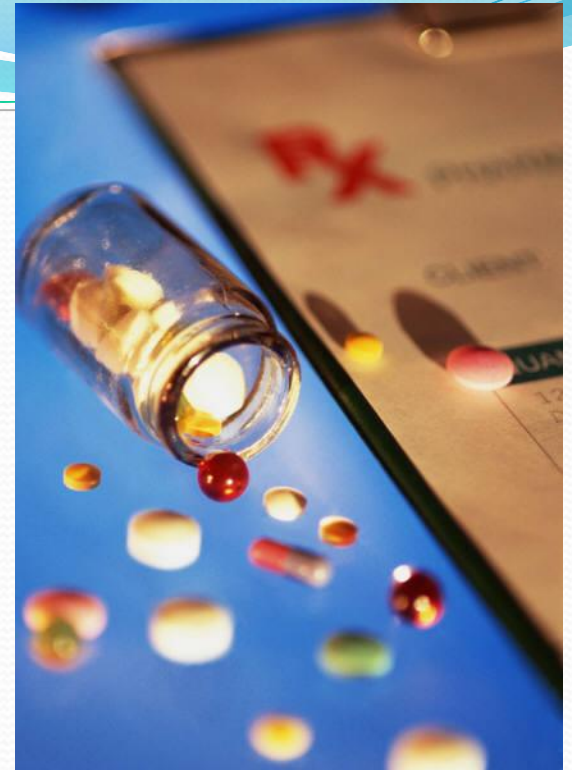
Benzodiazepines



- **Valium, Xanax, Ativan, Librium, Klonopin, Serax**
- **Dalmane, Halcion (used as sleep aides mostly-short term!!)**
- **Used for anxiety, panic disorders, ETOH withdrawal, muscle spasm, sedation, insomnia, and epileptics/seizures**
- **Use only short term because of dependency issues**
- **Avoid ETOH**
- **Causes sedation-don't drive!!**

Benzodiazepines

- **Side Effects;**
 - Drowsiness, confusion, sedation, and lethargy
- **Toxic Effects;**
 - Respiratory depression esp. with ETOH use!
- **Contraindications;**
 - Combination with other CNS depressants
 - Renal or hepatic dysfunction
 - History of drug abuse or addiction
 - Depression and suicidal tendencies
- **Teaching;**
 - Use short term due to drug dependency issues
 - Avoid ETOH and other CNS depressants
 - Can impair ability to drive
 - Do not use with someone who has a hx of drug dependency
 - D'C meds can cause withdrawal s/s



Nonbenzodiazepine Anxiolytic

- **BuSpar (Buspirone)=**
- reduces anxiety without strong sedative-hypnotic properties.
- Not a CNS depressant
- No potential for addiction
- Takes 2 weeks to feel effects



Nonbenzodiazepine Anxiolytic

- **Side Effects;**
 - Dizziness, dry mouth, nervousness, diarrhea, headache, excitement
- **Toxic Effects;**
 - Lethal dose is 160-550 times the daily recommended dose
- **Contraindications;**
 - Use with caution in PG women
 - Nursing mothers
 - Clients with renal or hepatic disease
 - Anyone taking MAOs
- **Teaching;**
 - Buspar is not associated with sedation, cognitive problems or withdrawal
 - Takes 2-4 weeks to feel effects
 - Some clients might feel restless, which could be incompleted anxiety



Sedative/Hypnotic Drugs

- Used to reduce anxiety and insomnia
- Can lead to tolerance and dependency
- Use short term

- **Drugs used**

benzodiazepines, i.e.
Dalmane, Restoril,
Halcion

Non-
benzodiazepines, i.e.
Ambien, Sonata,
Lunestra

Sedative/Hypnotic Benzodiazepine Teaching:

- Use short term(1-2 weeks)
- Carefully need to taper these off-never stop cold turkey
- Do not take with other meds without talking to provider first
- Do not drive if sedated on these!!

ADD/ADHD-Psychostimulants

- **Ritalin, Adderall, Dexedrine, Concerta, Focalin, Metadate, Methylin**
- **S/E:** wt. loss, anorexia, insomnia, headache, long-term growth suppression
- Potential for abuse
- **Action=**
increasing the release and blocking the reuptake of monoamines (dopamine, norepinephrine)
- Also used to treat narcolepsy

ADD/ADHD-Psychostimulants

- Intended effects:
 - Increased attention span & concentration
 - Decreased distractibility, hyperactivity, and impulsivity
 - Treatment of ADHD, ADD, & narcolepsy
- S/E:
 - Anorexia
 - Wt. loss
 - Growth retardation in children
 - Insomnia
 - Headache
 - Cardiovascular effects-high blood pressure, dysrhythmias
- Contraindications:
 - Hx of drug abuse & dependency, severe anxiety, anorexia, MAIOIs



ADD/ADHD- Non-Stimulants

- **Strattera** (atomoxetine)
 - Controls symptoms thru selective inhibition of norepinephrine
 - Takes 1-3 weeks to feel effects
 - No abuse potential and is not considered a controlled substance

Meds for Alzheimer's

- Drugs here are used to slow the progression of the disease
- **Memantine** (Namenda, Ebixa)
- **Cognex** (tacrine)
- **Aricept** (donepezil)
- **Exelon** (Rivastigmine)
- **Razadyne** (galantamine)



Herbal Medicines

- Ginkgo biloba-helps with memory
- Kava-Kava
- St. John's Wart



Thank you

- Any questions?